

KSP Datathon 2026 — Challenge 1

Intelligent Conversational AI & Crime Analytics Platform

EVALUATION CRITERIA MAPPING

Demo Coverage · Gap Analysis · Action List

Purpose	Map every problem-statement criterion to a demo beat, user story, or written submission section. Surface gaps before recording.
Modules covered	Modules 1–6, 9–10 (Modules 7–8 deferred to Phase 2)
Criteria checked	10 modules × individual sub-criteria + 5 top-level capability bullets = 42 criteria total
Status legend	STRONG (green) · COVERED (blue) · THIN (amber — needs attention) · GAP (red — must fix)
Output	Prioritised gap list with effort estimates and target (demo vs write-up vs narrator script)
Version	v1.0

1. How to Read This Document

The problem statement's ten modules and five capability bullets form the implicit scoring rubric. This document maps each criterion to its current coverage state, identifies what is missing from the demo, and produces a prioritised action list.

The core insight guiding this analysis: the demo currently shows Modules 1, 2, 5, 6 strongly. It underrepresents Modules 3 and 4, and it treats Modules 9 and 10 as a 20-second trust callout at the end rather than as cross-cutting properties visible throughout the demo. The fix is not more demo beats — it is weaving trust indicators into existing beats and adding one strong Module 4 image to the written submission.

Status	Meaning
STRONG	Criterion is explicitly demonstrated in a named demo beat with clear visual evidence
COVERED	Criterion is addressed in the demo or write-up — adequately but not prominently
THIN	Criterion is technically present but a jury member could miss it — needs a small fix
GAP	Criterion has no current demo or write-up coverage — must be addressed before submission

2. Module-by-Module Coverage

MODULE 1 — Conversational Crime Intelligence Interface

Criterion	Covered by	Demo moment	Status	Gap / Action
Natural language queries — English	Stories 1, 7, 16 · C1, C5	Beat 1 (English queries)	STRONG	✓ Covered
Natural language queries — Kannada	Stories 1, 16 · C1, C5	Not shown live	THIN	⚠ Add one Kannada-script typed query to Beat 1 Q1 — suspect name in Kannada opens the sequence
Retrieve FIRs, accused, victims, locations, status	Stories 1, 7, 18 · C6	Beat 1 response shows all fields	STRONG	✓ Covered
Context-aware multi-turn conversations	Story 1 · C1	Beat 1 — 3 queries, no context repeat	STRONG	✓ Explicitly demoed — strongest Module 1 moment
Save conversation as PDF	Story 6 · C3, C14	Trust callout — PDF button visible	THIN	⚠ Add one narrator line pointing to PDF button explicitly
Voice interaction — English + Kannada	Stories 1, 16 · C2	Not in demo	COVERED	✓ In write-up as 'designed — Kannada STT pipeline identified'. Voice latency risk justifies exclusion from live demo.

MODULE 2 — Criminal Network & Relationship Analysis

Criterion	Covered by	Demo moment	Status	Gap / Action
Links between accused, victims, locations	Story 4 · C9, C8	Beat 2 — graph materialises	STRONG	✓ Covered
Financial account links	Story 9 (Phase 2) · C9	Not in demo or write-up	COVERED	✓ Note Phase 2 deferral explicitly in submission — Module 7 section
Visualization of criminal networks	Story 4 · C9	Beat 2 — the graph	STRONG	✓ The visual centrepiece of the demo
Detection of organized crime groups	Story 4, 12 · C9, C8	Beat 2 — gang node present	THIN	⚠ Make gang node visually distinct (different icon/colour from person nodes) so 'organized crime detection' is visible without narration
Repeat offender networks	Story 12 · C8, C6	Beat 1 — history sheet returned	COVERED	✓ Story 12 (watchlist) referenced in write-up

MODULE 3 — Crime Pattern & Trend Analytics

Criterion	Covered by	Demo moment	Status	Gap / Action
Analysis across time and crime type	Stories 8, 10 · C10a, C6	Beat 3 — month-over-month by crime type	STRONG	✓ Covered
Analysis across geography	Story 10, 20 · C6, C10a	Beat 3 — single district only	THIN	⚠ Add Beat 3 follow-up: 'compare to neighbouring districts?' — adds geographic dimension without map, costs 5 sec

Criterion	Covered by	Demo moment	Status	Gap / Action
Crime hotspots and emerging clusters	Story 8 · C10a, C7	<i>Not demoed live</i>	COVERED	✓ In write-up: 'cluster surfacing runs MO match automatically across full dataset on a schedule'
Modus operandi trend analysis	Story 2 · C7, C12	<i>Beat 4 — 6 MO-matched cases across 4 stations</i>	STRONG	✓ Beat 4 is the strongest Module 3 demonstration
Seasonal / event-based trends	Story 21 · C10a, C6	<i>Not demoed</i>	COVERED	✓ In write-up: 'bandobast planning pack shows historical incident patterns for any event/date window'

MODULE 4 — Sociological Crime Insights

Criterion	Covered by	Demo moment	Status	Gap / Action
Crime patterns by demographic attributes	Story 13 · C17, C10	<i>Not demoed</i>	GAP	✗ Add static mockup screenshot in written submission: district-level crime vs. unemployment table for 5 Karnataka districts
Social risk factors influencing crime	Story 13, 24 · C17	<i>Not demoed</i>	GAP	✗ Covered by same mockup screenshot as above
Correlation with urbanisation, migration, economic stress	Story 13, 24 · C17	<i>Not demoed</i>	GAP	✗ One sentence in write-up: 'crime rate joinable with census indicators via district_socioeconomic table — literacy, unemployment, urbanisation, migrant %'

MODULE 4 NOTE Module 4 has zero demo moments in the current script. A static mockup screenshot showing a policymaker query result (correlation table) costs approximately 1 hour of design work and makes Module 4 visible to evaluators without requiring the full external data pipeline to be live.

MODULE 5 — Criminology-Based Offender Profiling

Criterion	Covered by	Demo moment	Status	Gap / Action
Identification of repeat / habitual offenders	Story 12 · C8, C6	<i>Beat 1 — history sheet (Rowdy, 3 cases) returned</i>	STRONG	✓ Covered in Beat 1
Behavioral analysis based on crime history and MO	Story 2, 12 · C7, C12	<i>Beat 4 — MO match with behavioral pattern across 6 cases</i>	STRONG	✓ Beat 4 is the strongest Module 5 moment
Risk scoring to prioritise investigation	Story 12 · C8	<i>Beat 1 — 'Risk level: High' visible</i>	THIN	⚠ Add narrator line: 'Rule-based scoring — case count, recency, conviction history — criteria shown on request.' Demonstrates contestability.

MODULE 6 — Investigator Decision Support

Criterion	Covered by	Demo moment	Status	Gap / Action
Automated case summaries and	Story 3, 19 · C12, C7	<i>Not demoed</i>	COVERED	✓ In write-up: 'chargesheet prep brief

Criterion	Covered by	Demo moment	Status	Gap / Action
investigation timelines				assembles timeline, evidence, sections automatically'
Similar past cases and investigation outcomes	Story 2 · C7, C12	Beat 4 — 6 similar cases with outcomes	STRONG	✓ Explicitly shown — the strongest Module 6 demonstration
Recommendations for investigative leads	Story 2 · C12	Beat 4 — 'common thread: pawnshop or CCTV within 48 hrs'	STRONG	⚠ Add one more line: 'Suggested next action: check pawnshops within 3km within 48 hours' — turns pattern insight into explicit lead recommendation

MODULES 7 & 8 — Financial Crime · Forecasting (Phase 2 — Deferred)

Module 7 — Financial Crime & Transaction Link Analysis	Deferred to Phase 2. No transaction tables in v1 schema. Note explicitly in submission with rationale: financial link analysis requires data sources beyond the FIR corpus.
Module 8 — Crime Forecasting & Early Warning	Deferred to Phase 2. C11 (predictive models) not built in v1. Note in submission: forecasting requires bias validation and historical data-quality work that should not be rushed.
Submission framing	Show both modules as labeled Phase 2 items on the roadmap slide. Framed as engineering judgment, not incompleteness.

MODULE 9 — Explainable AI & Transparent Analytics

Criterion	Covered by	Demo moment	Status	Gap / Action
AI responses supported by data references and evidence trails	C14 – citation layer	Trust callout — citation panel shown	THIN	⚠️ CRITICAL: Citations must be visible in EVERY beat as inline [1][2] superscripts. Trust callout shows what clicking one does. Not a 20-second segment — a cross-cutting property.
Visualization of reasoning paths and correlations	C14 – generated SQL visible	Trust callout — SQL shown collapsed	THIN	⚠️ Show generated SQL UNCOLLAPSED in at least one beat. Beat 3 (SP query) recommended — SQL is short and readable at 1080p.
Compliance with law enforcement accountability requirements	C16 – audit log	Trust callout — audit log panel	COVERED	✓ Covered. Mention DPDP Act 2023 in written submission (see Module 10).

MODULE 9 CRITICAL NOTE This module is the most underrepresented relative to its importance. The fix is not adding demo time — it is making citations an always-on UI element visible in Beats 1, 2, 3, and 4, not just the trust callout. Every chat bubble should show [1][2] superscripts from the very first response.

MODULE 10 — Secure Role-Based Access & Governance

Criterion	Covered by	Demo moment	Status	Gap / Action
Role-based access for all persona types	Story 14 · C15	Trust callout — role indicator; 3 persona switches in demo	STRONG	✓ Three persona switches make RBAC visible without narration
Secure handling with audit logs and traceability	Story 15 · C16	Trust callout — audit log panel	COVERED	✓ Covered. Log panel shown in trust callout.
Compliance with data protection frameworks	Stories 14, 15, 26 · C15, C16	Trust callout only	THIN	⚠️ Add one paragraph in written submission mentioning DPDP Act 2023: victim PII inaccessible to policymakers; all access logged per Section 8 obligations.

3. Five Top-Level Capability Bullets

These five bullets appear in the problem statement preamble and are likely used as a top-level scoring rubric before the ten modules. Every one must have either a demo moment or a prominent written-submission reference.

Criterion	Covered by	Demo moment	Status	Gap / Action
Crime pattern discovery	Stories 8,10,20 · Beat 3	Beat 3 — month-over-month anomaly	COVERED	⚠️ Add 'emerging cluster' framing to Beat 3 narrator line: '...and surfaces patterns before the monthly review'
Criminal network analysis	Story 4 · Beat 2	Beat 2 — graph	STRONG	✓ The visual centrepiece — strongest capability demonstration
Socio-demographic crime insights	Story 13 · Write-up only	No demo moment	GAP	✗ Add Module 4 mockup screenshot to written submission (same fix as Module 4)
Behavioral and criminological profiling	Stories 2,5,12 · Beat 4	Beat 4 — MO match + outcome analysis	STRONG	✓ Covered — Beat 4 is explicitly this capability
Proactive crime prevention intelligence	Stories 8,12,17 · Write-up	Not framed explicitly in demo	THIN	⚠️ Add to closing narrator: '...patterns surface before the monthly review — early enough to act on'

4. Consolidated Gap List — Prioritised Action Items

All gaps from Sections 2 and 3 consolidated and prioritised. Complete MUST-FIX items before recording. Complete SHOULD-FIX items before submission. NICE-TO-HAVE if time permits.

MUST-FIX — Complete before recording

Priority	Action	Where	Effort	Module(s)
MUST-FIX	Add inline citation superscripts [1] [2] to every system response in the demo. Citations are not a trust callout — they are always-on UI primitives visible from the first query.	UI build — chat bubble component	~1 hr	Module 9
MUST-FIX	Show generated SQL uncollapsed in Beat 3 (SP review query). SQL is short enough to read at 1080p and directly demonstrates transparent reasoning.	Demo recording — Beat 3	5 min	Module 9
MUST-FIX	Add one Kannada-script typed query to Beat 1 Q1 — suspect's name in Kannada as the opening query. Demonstrates multilingual capability without voice latency risk.	Demo recording — Beat 1	10 min	Module 1
MUST-FIX	Make gang node	UI build — graph	30 min	Module 2

Priority	Action	Where	Effort	Module(s)
	visually distinct in Beat 2 graph — different icon shape or colour from person nodes. Makes 'organized crime detection' visible without narration.	<i>renderer</i>		

SHOULD-FIX — Complete before submission

Priority	Action	Where	Effort	Module(s)
SHOULD-FIX	Add a static mockup screenshot to written submission showing a policymaker query: district-level crime rates vs. unemployment rates for 5 Karnataka districts. Covers Module 4 and the socio-demographic capability bullet.	<i>Written submission design</i>	~1 hr	Module 4 · Cap. bullet 3
SHOULD-FIX	Add 'Suggested next action: check pawnshops within 3km within 48 hours' as a final line in Beat 4 system response. Converts pattern insight into explicit lead recommendation.	<i>Demo script — Beat 4</i>	5 min	Module 6
SHOULD-FIX	Add narrator line in Beat 1 about risk scoring: 'Rule-based scoring — case count, recency, conviction history. Criteria shown on request.' Demonstrates contestability and ethical positioning.	<i>Narrator script — Beat 1</i>	5 min	Module 5
SHOULD-FIX	Add one paragraph to written submission on DPDP Act 2023: victim PII inaccessible to policymakers; all access logged per Section 8 obligations.	<i>Written submission</i>	30 min	Module 10
SHOULD-FIX	Add 'emerging cluster' framing to Beat 3 narrator: '...surfaces patterns before the monthly review — early enough to act on.'	<i>Narrator script — Beat 3</i>	5 min	Module 3 · Cap. bullet 5
SHOULD-FIX	Add one sentence to narrator script explicitly pointing to PDF export button in trust callout.	<i>Narrator script — trust callout</i>	2 min	Module 1

NICE-TO-HAVE — If time permits

Priority	Action	Where	Effort	Module(s)
NICE-TO-HAVE	Beat 3 follow-up: after the Mysuru anomaly result, add 'How does this compare to neighbouring districts?' Shows geographic comparison without a map.	Demo recording — Beat 3	15 min	Module 3
NICE-TO-HAVE	Voice demo: if Kannada STT is tested and reliable, add a 5-second voice query cutaway in Beat 1 or Beat 2. Only include if zero latency or zero error rate confirmed.	Demo recording	Varies	Module 1
NICE-TO-HAVE	Beat 4 closing: add a 'Subscribe to alert when this MO is registered again' button visible in the UI — demonstrates Story 17 (watchlist subscription) without a separate demo beat.	UI build	~1 hr	Module 6

5. Module Coverage Summary

Estimated evaluator visibility of each module, before and after the gap fixes. Scores are qualitative — 1 (no coverage) to 5 (explicitly and memorably demonstrated).

Module	Current score → After gap fixes
Module 1 — Conversational Interface	4 / 5 → 5 / 5 (Kannada query + PDF callout)
Module 2 — Criminal Networks	5 / 5 → 5 / 5 (gang node fix is cosmetic)
Module 3 — Crime Patterns	3 / 5 → 4 / 5 (geographic comparison + cluster framing)
Module 4 — Sociological Insights	1 / 5 → 3 / 5 (mockup screenshot in submission)
Module 5 — Offender Profiling	4 / 5 → 5 / 5 (risk scoring narration)
Module 6 — Decision Support	4 / 5 → 5 / 5 (lead recommendation line)
Module 7 — Financial Crime	N/A — Phase 2 (documented in roadmap)
Module 8 — Forecasting	N/A — Phase 2 (documented in roadmap)
Module 9 — Explainable AI	2 / 5 → 5 / 5 (inline citations + SQL visible — BIGGEST FIX)
Module 10 — RBAC & Governance	4 / 5 → 5 / 5 (DPDP Act paragraph in submission)
Top-level capability bullets	3 / 5 covered strongly → 5 / 5 (Module 4 screenshot + prevention framing)

KEY INSIGHT *Module 9 (Explainable AI) has the largest gap relative to its importance. It is the evaluator's most likely differentiator between teams — and it requires only UI work (inline citations), not new features. Every team will claim explainability. Only teams that show it in every beat will be believed.*